



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001:2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Computer Science and Business Systems

Academic Year 2023– 2024 (Odd/ Even Semester)

Degree, Semester & Branch: B.Tech, IV & CSBS

Course Code & Title: AL3452 & Operating Systems

Name of the Faculty member(s): Dr.M.Gomathy Nayagam

Innovative Practice Description

- **Unit/Topic:**Unit II/ CPU Scheduling
- **Course Outcome:** CO2
- **Topic Learning Outcome:** UO2a
- **Activity Chosen:** VLAB
- **Justification:**

Visualization is a teaching strategy which refers to the practical visualization of the usage of CPU scheduling will improve the understanding of the students on the concept. Theoretical explanation of concepts is not sufficient and also not understandable one for beginners. Hence, I chose VLAB for explaining the aforementioned concept.

- **Time Allotted for the Activity:**30 Minutes
- **Details of the Implementation:**

- Material used :

- Virtual Lab - <https://naim30.github.io/OS-virtual-lab/schedulingAlgo/schedulingAlgo.html>
- Prerequisite : Knowledge on the CPU Scheduling. (Obtained by Lecture)
- Visualization of CPU scheduling is shown with the help of virtual lab.
- At each step, students are asked to perform on their own. The answers are verified in the virtual lab.
- Students also motivated to execute it for more expressions in their laptop or in their lab session.

- **CO–PO/ PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO 3	3	3	3	-	-	-	-	1	1	1	-	1	1	1	-

(1– Low

2–Moderate

3 – High)

• **PO/ PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO8	PO9
	3	3	3	1	1
Justification for correlation	The students could be able to apply the basic engineering knowledge to use the various page CPU Scheduling	The students could be able use the various CPU Scheduling with different case studies.	The students could able to provide the solution for identifying the relationship between various CPU Scheduling	The students could be able to solve the problem without copying from others during the event.	The students can be able to discuss with their course instructor and solve their problem without any hesitation.

Innovative practice	PO10	PO12	PSO1	PSO2
	1	1	1	1
Justification for correlation	The students can be able to apply and solve the problem individually.	The students could be able to continuously learn about the Scheduling algorithms	The students could be able to apply the various CPU Scheduling during OS usage and iOS for the various services of OS	The students could be able to apply the various CPU Scheduling during OS usage and iOS for the various services of OS

- **Images/Screen shot of the practice:**

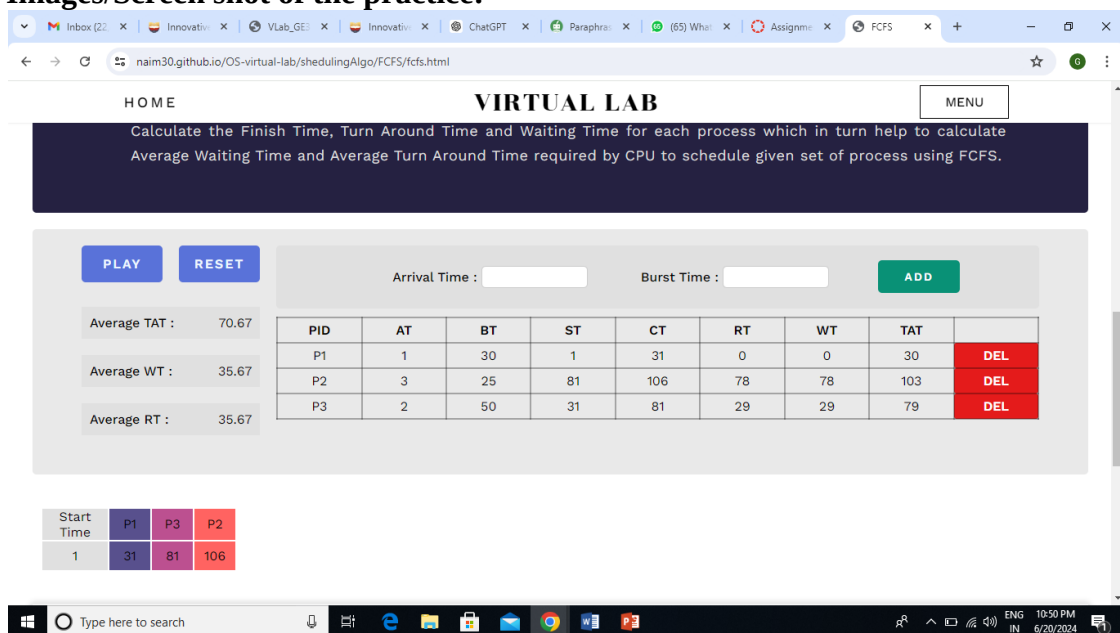


Fig. Glimpse of VLAB Platform

- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

Students felt that Visualization on the topic gave them better understanding on the topic very easily. At Any time, students can verify their answers with the help of virtual lab.

- ❖ **Benefit of the Practice (E.g. Outcome Attainment would have increased due to innovative practice over conventional practice)**

- Flipped mode of teaching attracted the student's attention. They gave better focus on the topic than conventional teaching method.

- ❖ **Challenges faced in implementation:**

- It is not a student centered method. So, few students not actively participate in this type of learning.

References:

- ❖ . <https://www.ritrjpm.ac.in/images/computer-science/CS6801-MCAP-MGM.pdf>
- ❖ https://www.ritrjpm.ac.in/images/computer-science/TPS_Demo_SM.pdf
- ❖ https://www.netdevgroup.com/online/content/vmita/ictacademy.html?utm_source=VMware&utm_campaign=ict#learn

Signature of Faculty Member
(Dr.M.Gomathy Nayagam)

HOD